Recap – dealing with VLE

Binary mixtures have 3D phase diagrams

Reduced phase diagrams are 2D slices through 3D diagram

$$\gamma_i x_i f_i^l = \overline{f}_i^v = y_i f_i^v$$

$$\gamma_i x_i P_i^{SAT} = y_i P$$

Raoult's law = Ideal Mixtures at low pressure

Large deviations from Raoult's law can result in an azeotrope = a minimum or maximum in $P - x_1$ with $x_1 = y_1$

An example of the experimental determination of activity coefficients

n-pentane (1) and propanal (2) form an azeotropic mixture at T = 40°C, P = 1.364 bar and $x_1 = 68.2\%$

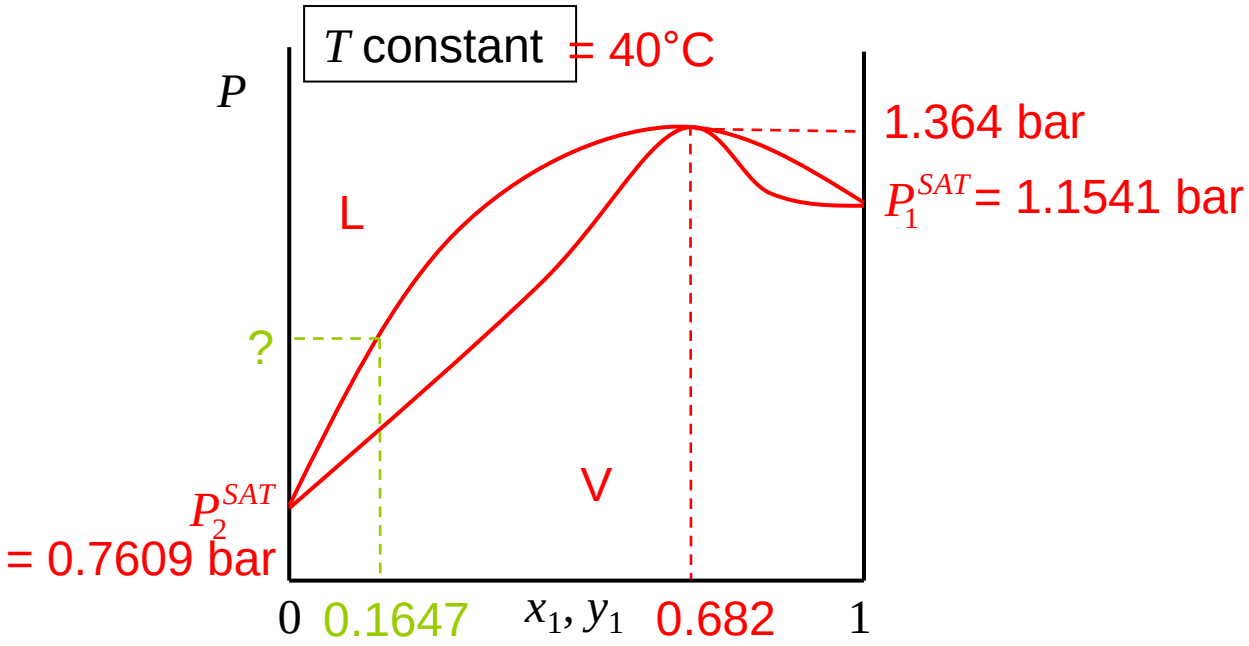
Use the van Laar model to estimate the vapour pressure for x_1 = 16.47% and T = 40°C

Compare with the measured value of 1.1622 bar. Suggest reasons for the discrepancy

Data:

n-pentane: $P_1^{SAT} = 1.1541\,\text{bar}$ propanal: $P_2^{SAT} = 0.7609\,\text{bar}$

Solution:


T constant = 40°C
P
1.364 bar
P_1^{SAT} = 1.1541 bar
L
?
V
P_2^{SAT}
= 0.7609 bar
0
0.1647
x_1, y_1
0.682
1

Start from $\gamma_i x_i P_i^{SAT} = y_i P$

Azeotrope $\Rightarrow x_1 = y_1$: $\gamma_1 = \dfrac{\cancel{y_1} P}{\cancel{x_1} P_1^{SAT}} = 1.182$

$\gamma_2 = \dfrac{\cancel{y_2} P}{\cancel{x_2} P_2^{SAT}} = 1.793$

at azeotrope

Van Laar (see handout 5):

$$\ln \gamma_1 = \frac{\alpha}{\left(1 + \dfrac{\alpha x_1}{\beta x_2}\right)^2} \qquad \ln \gamma_2 = \frac{\beta}{\left(1 + \dfrac{\beta x_2}{\alpha x_1}\right)^2}$$

2 parameters, 2 unknowns, so solve

Rearrange Van Laar: $\alpha = \left(1 + \dfrac{x_2}{x_1}\dfrac{\ln \gamma_2}{\ln \gamma_1}\right)^2 \ln \gamma_1 = 1.1547$

$\beta = \left(1 + \dfrac{x_1}{x_2}\dfrac{\ln \gamma_1}{\ln \gamma_2}\right)^2 \ln \gamma_2 = 1.5220$

Using these parameters, substitute $x_1 = 0.1647$ into Van Laar eqs:

T = 40˚C again (remember α,β are T-dependent)

$$\ln\gamma_1 = \frac{\alpha}{\left(1+\frac{\alpha x_1}{\beta(1-x_1)}\right)^2} \Rightarrow \gamma_1 = 2.395$$

and similarly $\gamma_2 = 1.026$

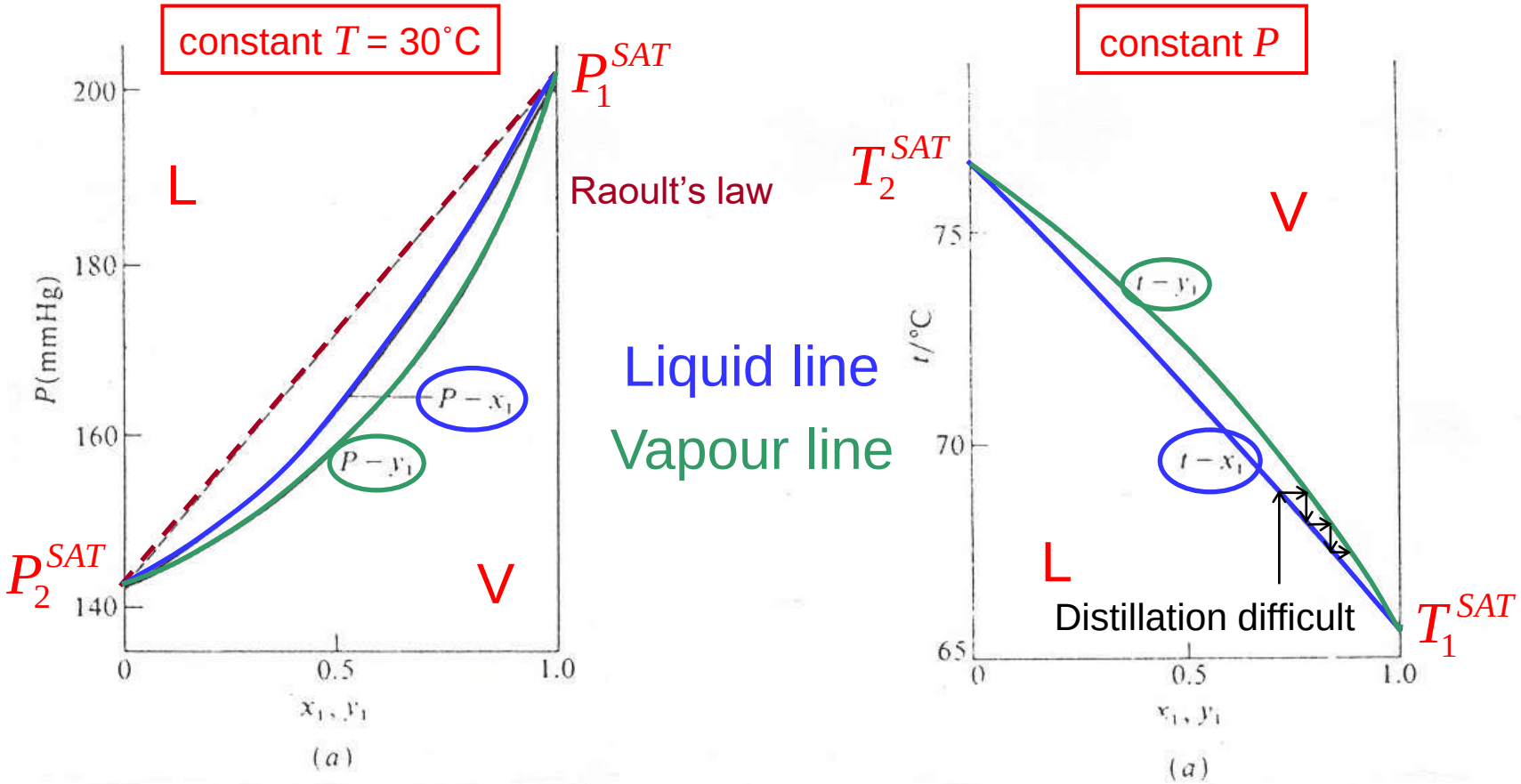
Now $$P = x_1\gamma_1 P_1^{SAT} + x_2\gamma_2 P_2^{SAT} = 1.107 \text{ bar}$$

$$\frac{\Delta P}{P_{MAX} - P_{MIN}} = \frac{(1.107 - 1.1622)}{(1.364 - 0.7609)} = 9.1\%$$

α,β from one point
propanal fairly polar $\Rightarrow$ answer not v. accurate

Some examples of the behaviour of binary mixtures

Moderate negative deviation from Raoult's law – due to H-bonding

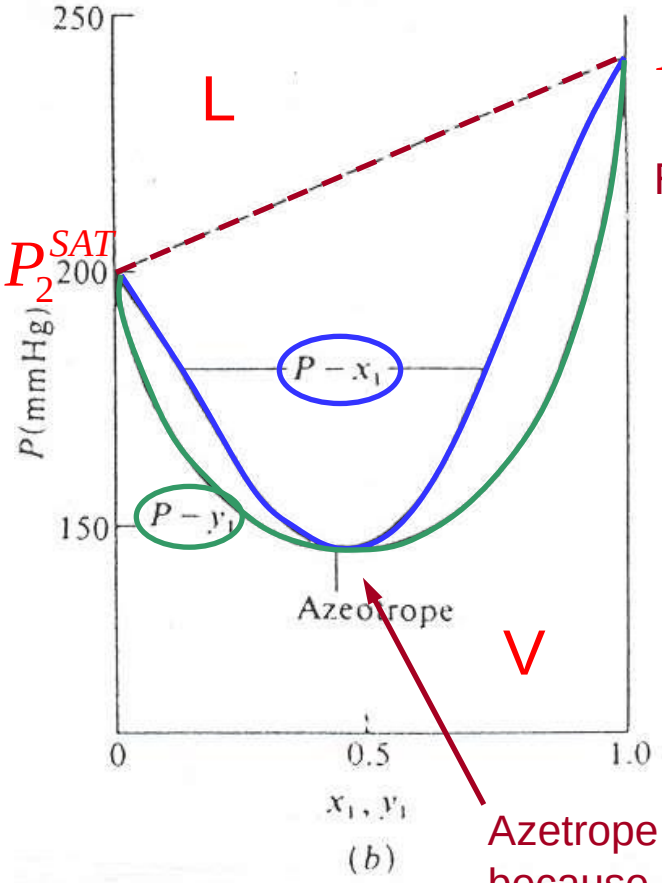


(a) tetrahydrofuran (1) / carbon tetrachloride (2) at 30°C

[Diagram taken from Smith and Van Ness, "Introduction to Chemical Engineering Thermodynamics", MacGraw Hill.]

Large negative deviation from Raoult's law – due to H-bonding – resulting in a minimum boiling pressure azeotrope

constant T = 30˚C

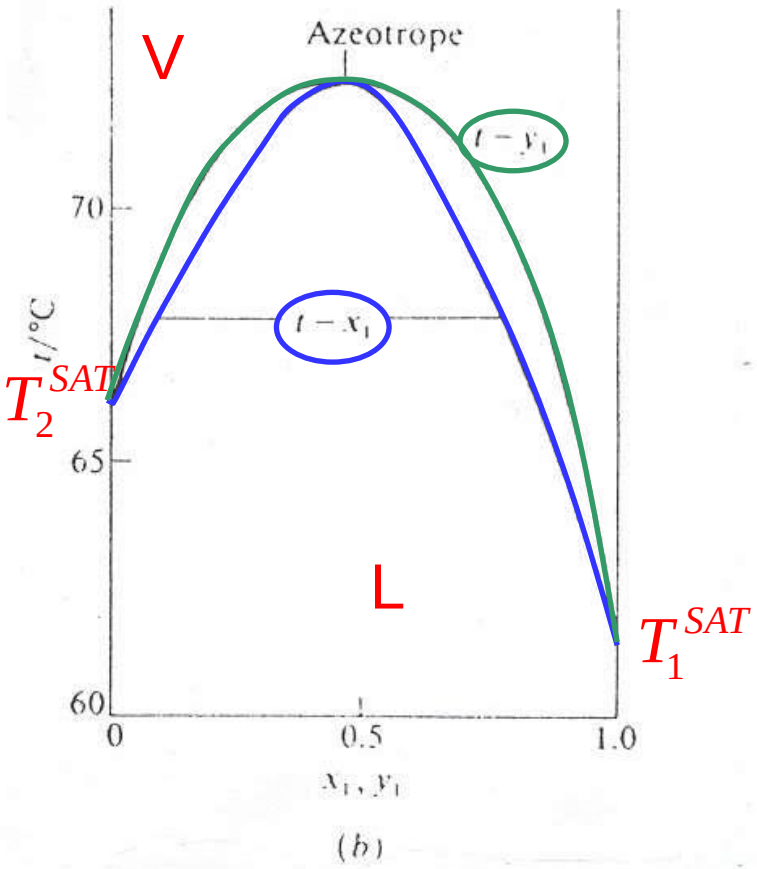


Raoult's law

Liquid line

Vapour line

constant P

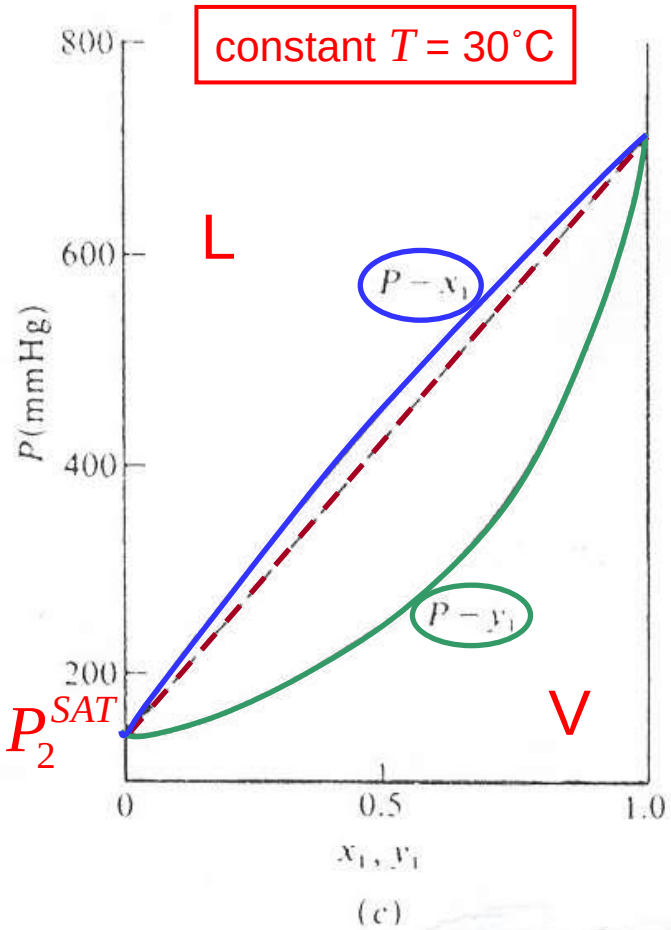


Azetrope at approx x_1 = 0.5 because 1:1 H-bonding

(b) chloroform (1) / tetrahydrofuran (2) at 30°C

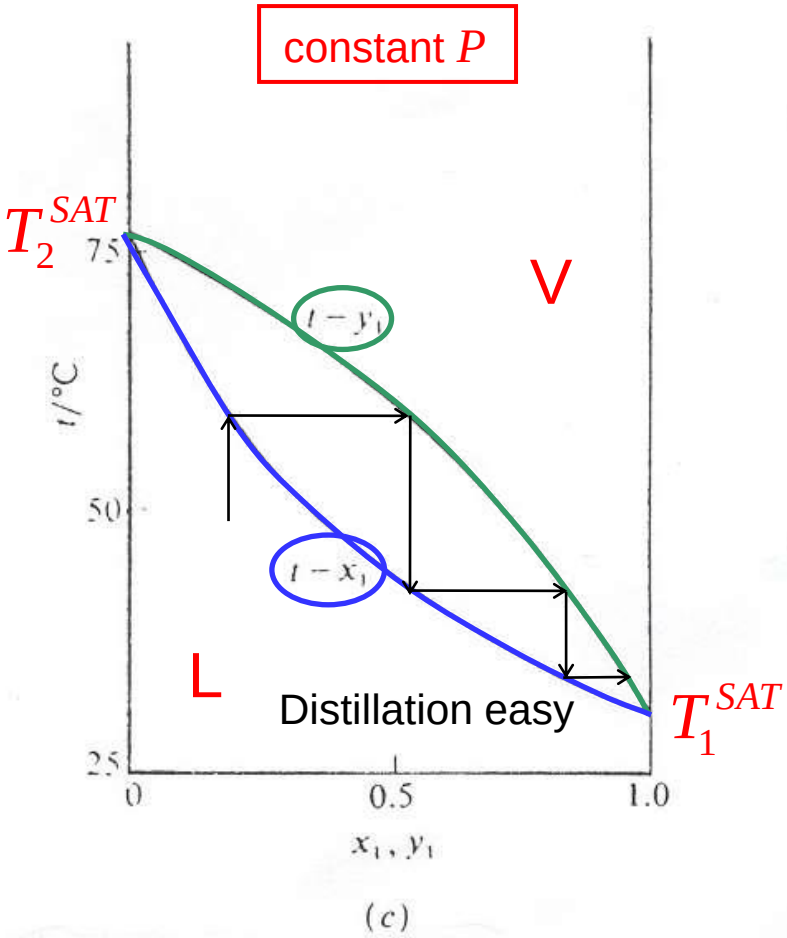
[Diagram taken from Smith and Van Ness, "Introduction to Chemical Engineering Thermodynamics", MacGraw Hill.]

Moderate positive deviation from Raoult's law – due to poor H-bonding



Liquid line

Vapour line

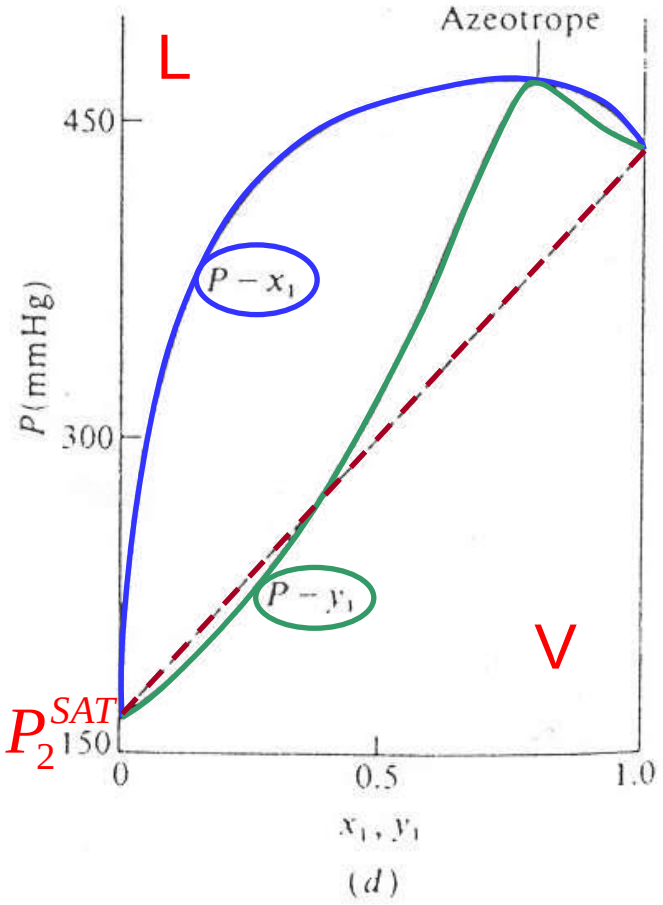


(c) furan (1) / carbon tetrachloride (2) at 30°C

[Diagram taken from Smith and Van Ness, "Introduction to Chemical Engineering Thermodynamics", MacGraw Hill.]

Large positive deviation from Raoult's law – because toluene disrupts H-bonding of ethanol – resulting in a maximum boiling pressure azeotrope

constant T = 65˚C



P_1^{SAT}

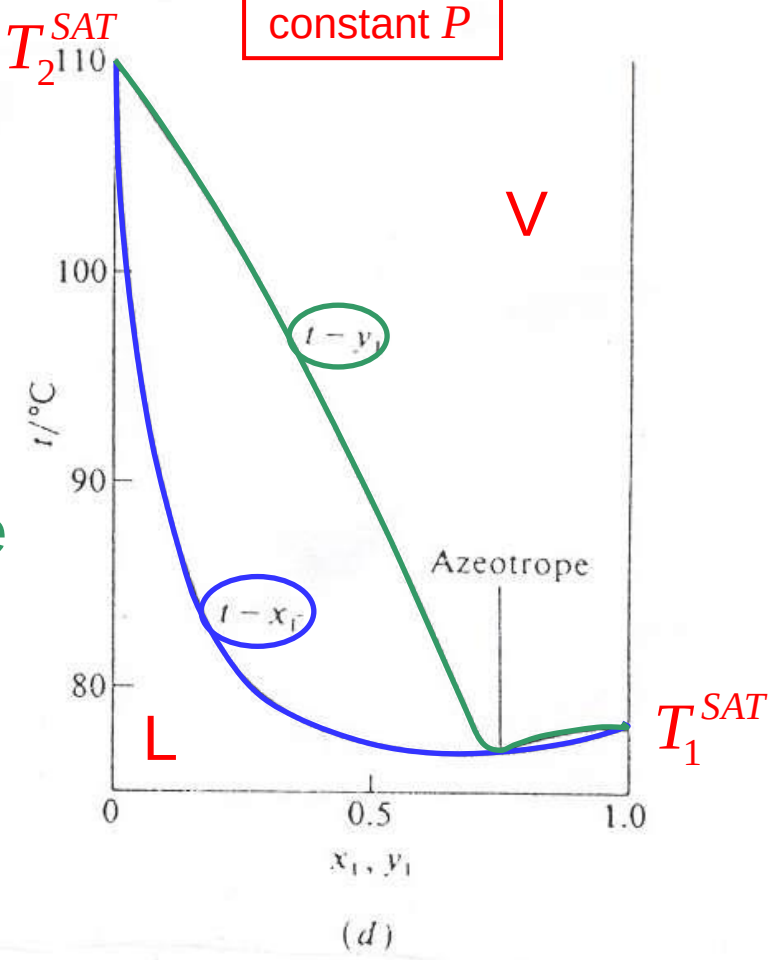
Raoult's law

Liquid line

Vapour line

P_2^{SAT}

constant P



T_1^{SAT}

(d) ethanol (1) / toluene (2) at 65°C

[Diagram taken from Smith and Van Ness, "Introduction to Chemical Engineering Thermodynamics", MacGraw Hill.]

Solubility of Gases in Liquids

From a thermodynamic point of view, there is no difference between dissolved gases and any other VLE, so:

$$\mu_i^l = \mu_i^v$$

or $$\overline{f}_i^l = \overline{f}_i^v$$

still holds

Reliable EOS's are rarely available, so we resort to

$$\gamma_i x_i f_i^l = \overline{f}_i^v \approx y_i f_i^v \qquad \text{(approx } \gamma_i x_i P_i^{SAT} = y_i P \text{)}$$

fugacity of the pure liquid at same T,P as the mixture

Above the critical temperature, the pure liquid does not exist

Thus, for $T > T_C$, f_i^l is undefined

There are two ways of proceeding for such cases:

(a) Extrapolation to hypothetical liquid phase

Recall that:

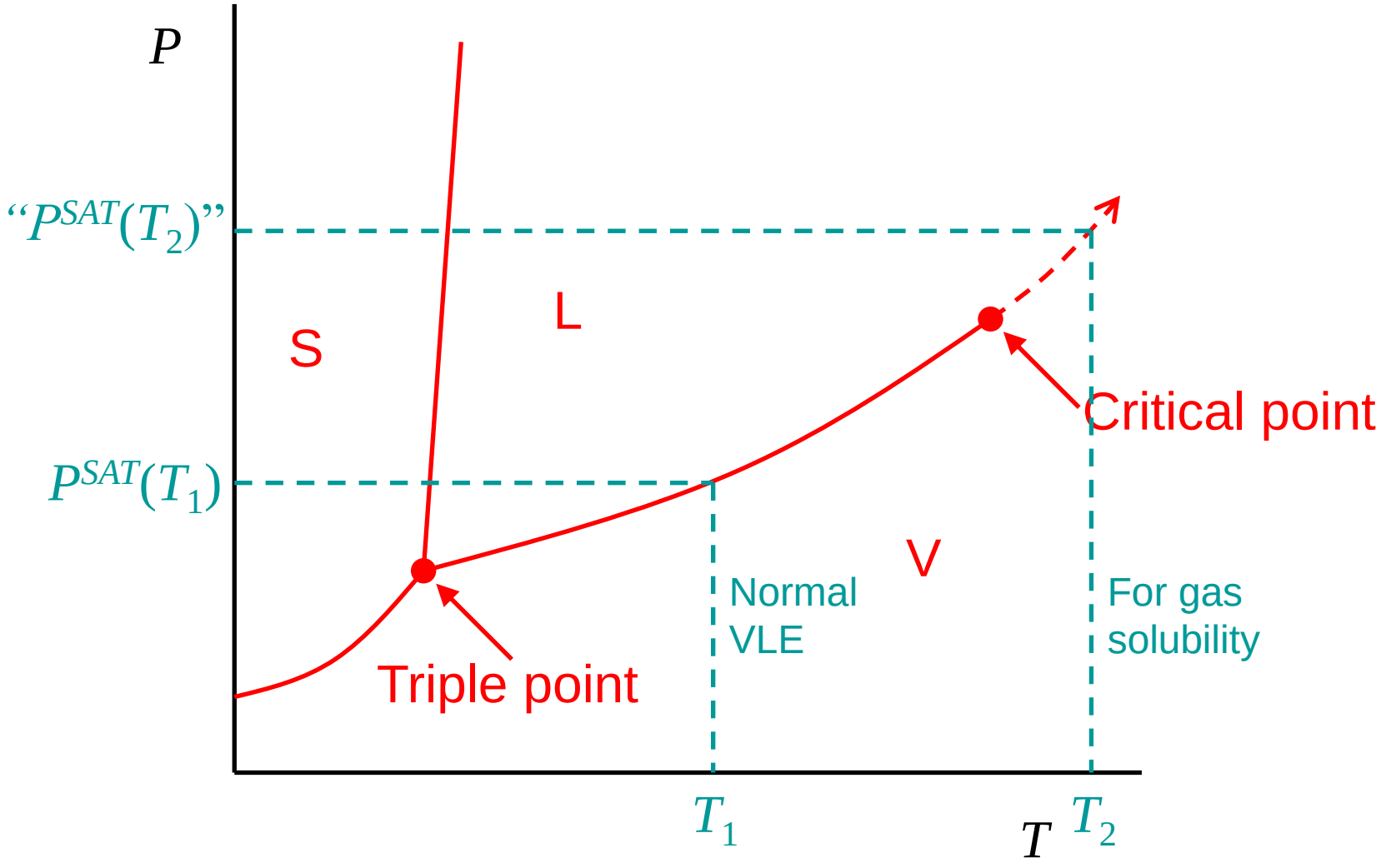
$$f_i^l = P_i^{SAT}\phi_i^{SAT}\exp\left(\frac{1}{RT}\underline{V}_i^l(P - P^{SAT}\right)$$

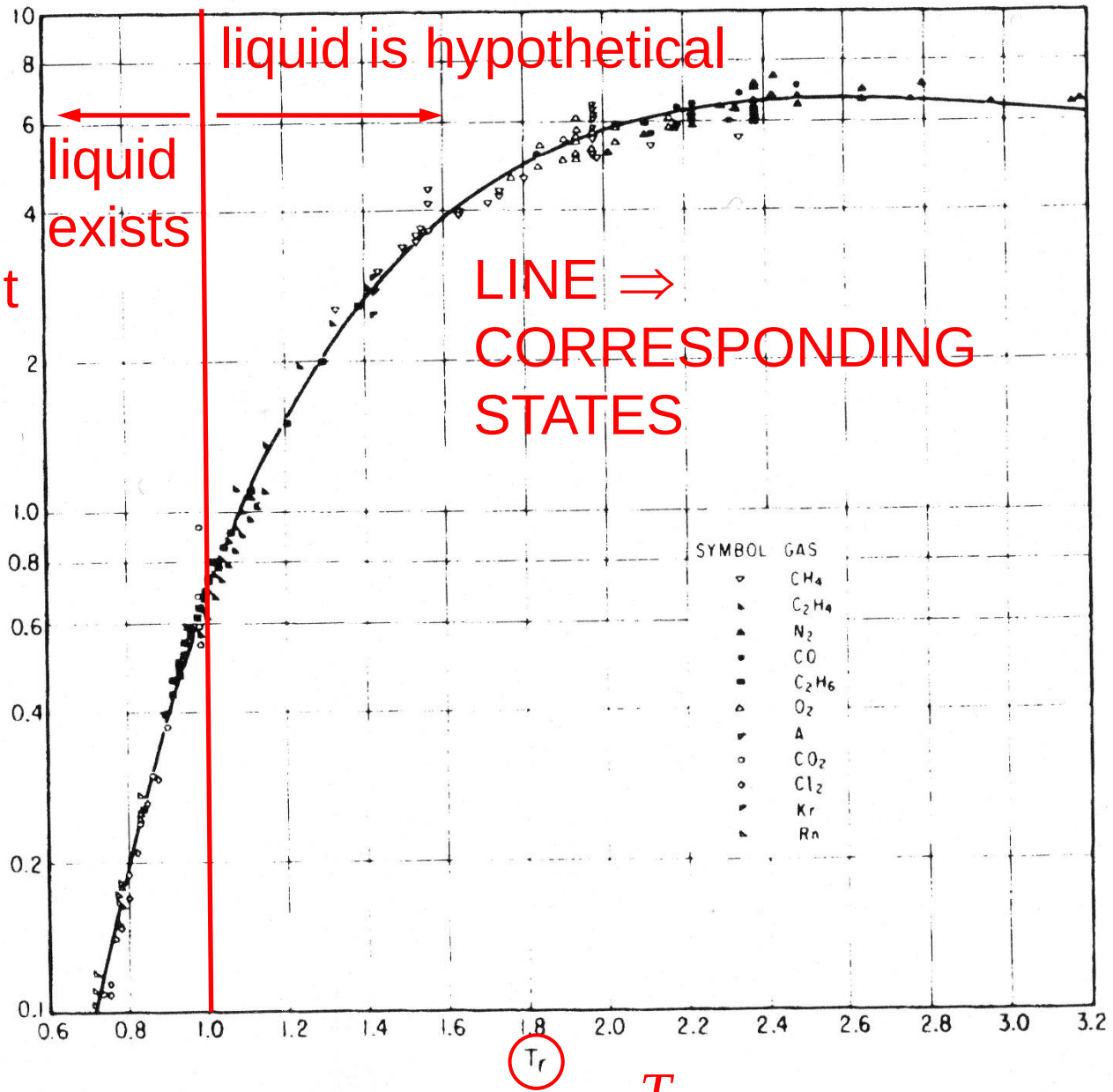
We may often neglect the Poynting correction and also assume that $\phi_i^{SAT} \approx 1$ (accurate at low *P*), so that $f_i^l \approx P_i^{SAT}$

We may therefore estimate f_i^l

by extrapolating the saturation line to the temperature of interest:

Pure substance phase diagram (schematic)





f_i^l is what we want

$\frac{f^L}{P_c}$

critical pressure of dissolved gas

$$T_r = \frac{T}{T_C}$$ ← of dissolved gas

[Diagram taken from Prausnitz and Shair (AIChEJ 7 [1961])]

Ideal Solubility

is defined as the solubility in an ideal solution, so $\gamma_i = 1$

Thus

fugacity of the gas $\approx y_i P$

$$x_i^{ideal} f_i^{l} = \overline{f}_i^{v}$$

fugacity of the pure liquid at same T,P as the mixture

$$x_i^{ideal} = \frac{\overline{f}_i^{v}}{f_i^{l}} = \frac{\overline{f}_i^{l}}{f_i^{l}}$$

We can only calculate the ideal solubility if we have a value for the hypothetical pure liquid fugacity

The ideal solubility provides a basis of comparison for real behaviour

It is independent of the solvent

Example

The following data have been reported for the solubility of methane in various solvents at T = 25°C, P = 10.13 bar, y_{CH_4}= 0.1 (the remainder of the gas is nitrogen):

Solvent	x_{CH_4}
benzene	2.07×10^{-3}
carbon tetrachloride	2.86×10^{-3}
cyclohexane	2.83×10^{-3}

(a) Calculate the ideal solubility of CH_4 under these conditions and compare with the above values.

Other data:

N_2: T_C = 126.2K, P_C = 33.9 bar

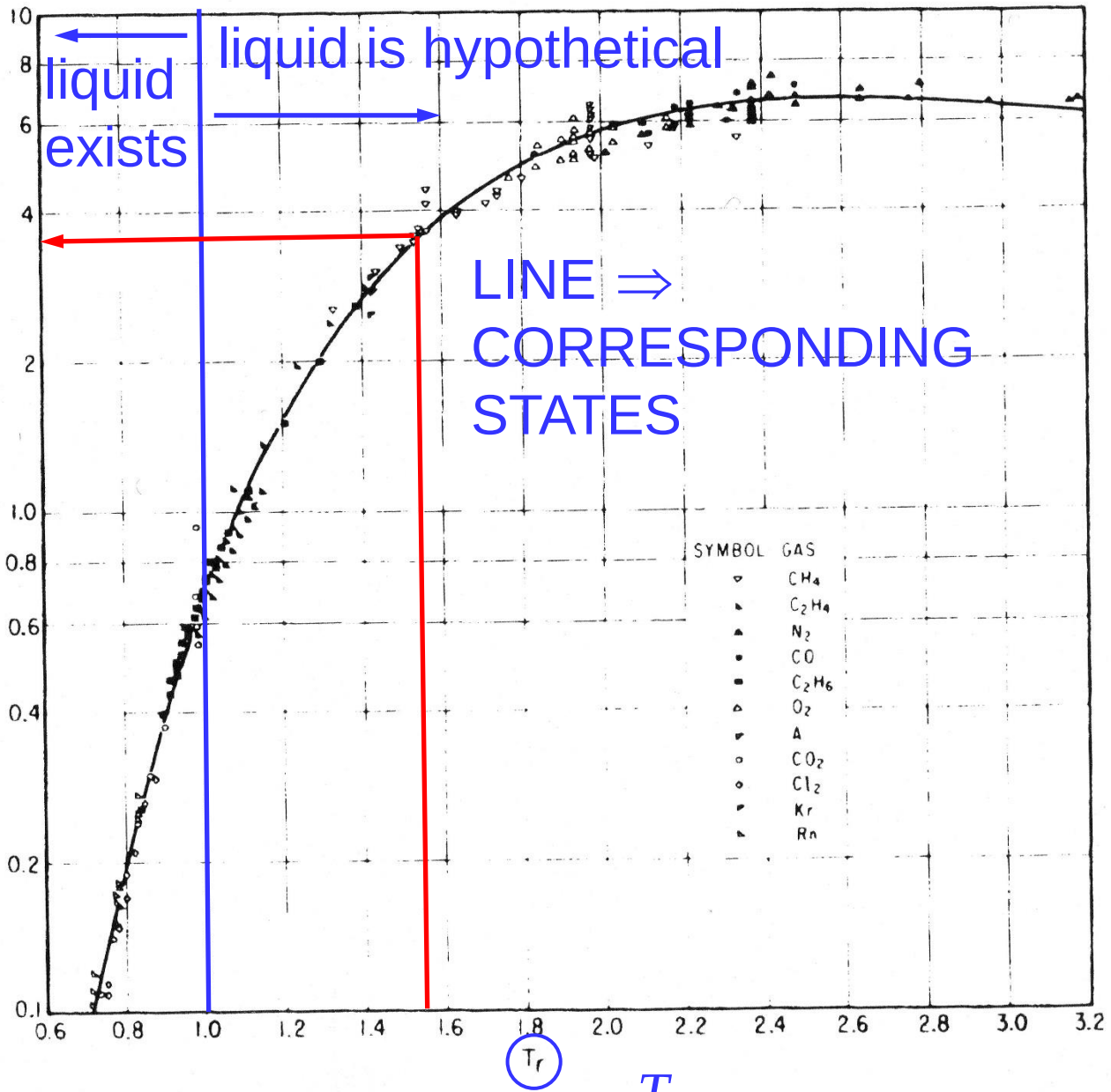
CH_4: T_C = 190.6K, P_C = 46.0 bar

Solution

$T >> T_C$ and $P << P_C$ for both N_2 and CH_4, so assume IGs

$$x_{CH_4}^{ideal} = \frac{y_{CH_4} P}{f_{CH_4}^{SAT}} \leftarrow \overline{f}_{CH_4}^{v} \text{ assuming IG}$$

Use Prausnitz and Shair correlation, with $\frac{T}{T_C} = \frac{298.2}{190.6} = 1.56$



$$\frac{f_{CH_4}^{SAT}}{P_C} = 3.65$$

critical pressure of dissolved gas

$$T_r = \frac{T}{T_C}$$

of dissolved gas

[Diagram taken from Prausnitz and Shair (AIChEJ 7 [1961])]

Use Prausnitz and Shair correlation, with $\frac{T}{T_C} = \frac{298.2}{190.6} = 1.56$

$$\Rightarrow f_{\mathrm{CH_4}}^{SAT} = 3.65 P_C = 167.9\ \mathrm{bar} \Rightarrow x_{\mathrm{CH_4}} = 6.03 \times 10^{-3}$$

too large, so solubility not ideal

(b) Calculate the activity coefficient of methane in these solvents

Solution

real partial pressure

$$\gamma_i = \frac{y_{CH_4} P}{x_{CH_4} f_{CH_4}^{SAT}} = \frac{x_{CH_4}^{ideal}}{x_{CH_4}}$$

ideal solubility

real solubility

ideal partial pressure

Solvent	γ_{CH_4}
benzene	2.91
carbon tetrachloride	2.11
cyclohexane	2.13

(b) Use of Henry's Law

The liquid phase concentration, x_i, of a sparingly soluble gas (i.e. one for which $x_i \ll 1$) is linearly proportional to its gas phase fugacity (or pressure if the gas behaves ideally)

i.e. $\overline{f}_i^{\,v} = \overline{f}_i^{\,l} = x_i H_i(T)$

$H_i = \gamma_i^{\infty} f_i^{l}$ — at infinite dilution; hypothetical

$H_i(T)$ is Henry's constant

Note that it has the same units as fugacity or pressure and is a function of temperature and both the gas and the solvent used.

We can go beyond the limiting “linear” realm of Henry’s law by using $H_i(T)$ to define a new standard state.

The conventional definition of activity is:

$$a_i = x_i \gamma_i = \frac{\overline{f}_i^{\,l}}{f_i^{\,l}}$$ i.e. the activity is unity for the pure liquid

We can define a new activity coefficient, γ_i^*, with Henry’s constant as the standard state fugacity

$$a_i^* = x_i \gamma_i^* = \frac{\overline{f}_i^{\,l}}{H_i}$$

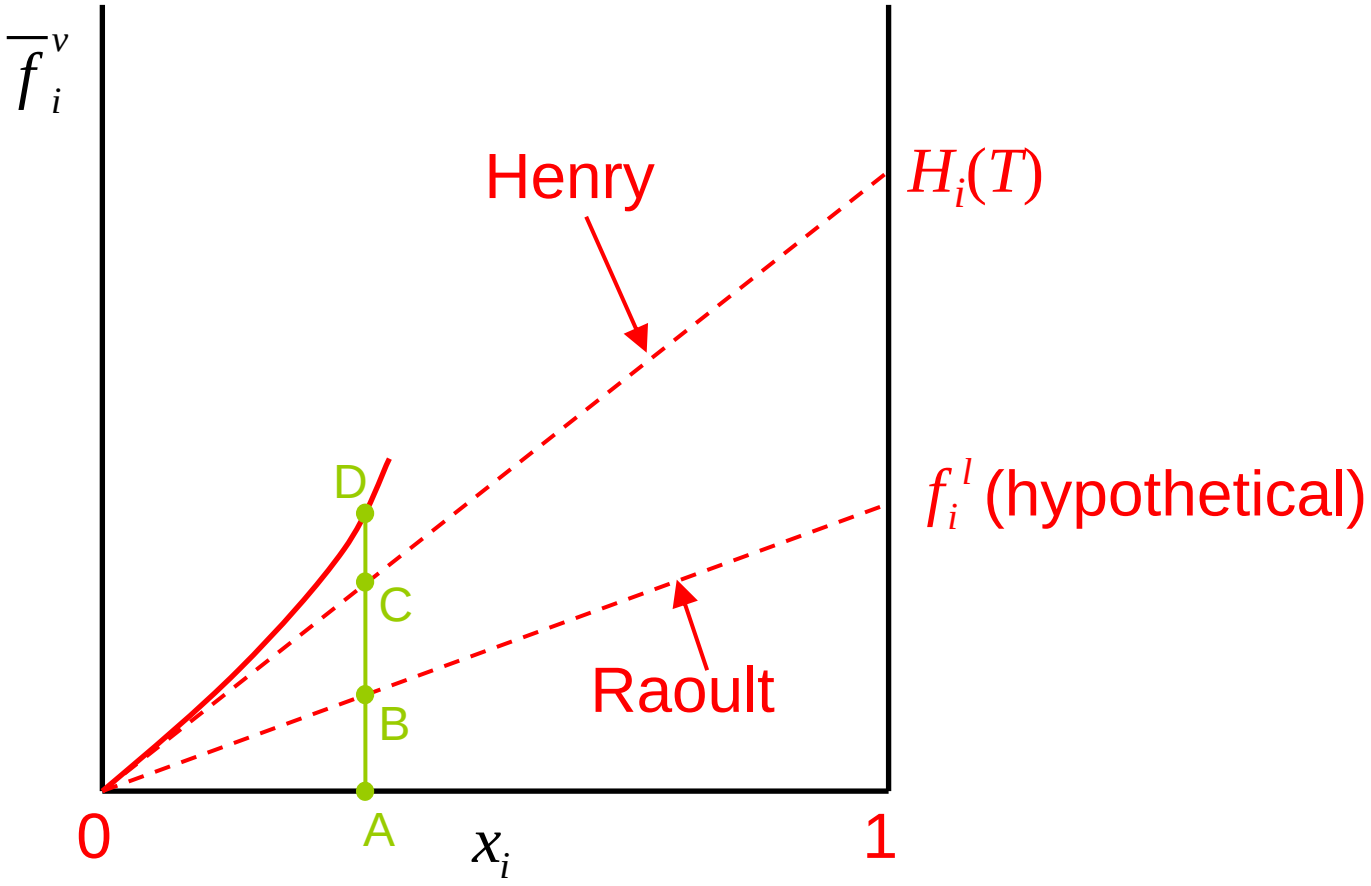
The justification for this is convenience
(we don’t need to know the hypothetical $f_i^{\,l}$)

We can choose any standard state,
so long we are consistent

The new, general equation for solubility is:

$$\gamma_i^* x_i H_i = \overline{f}_i^v$$

This is best understood graphically



$$\gamma_i^* = \frac{\overline{f}_i^{\,v}}{x_i H_i} = \frac{\mathrm{AD}}{\mathrm{AC}}$$

$$\gamma_i = \frac{\overline{f}_i^{\,v}}{x_i f_i^{\,l}} = \frac{\mathrm{AD}}{\mathrm{AB}}$$